

Data Task for research associate position with PER at Columbia University

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Code revisions and grammatical checks are performed using ChatGPT; however, the core ideas are original, drawn from my own insights, and supported by academic resources.

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Chapter 1

Summary Statistics

1.1 Summary Statistics Table

As shown in Table 1.1, the summary statistics for the commuting variables are presented below.

Table 1.1: Summary Statistics of Commuting Variables

Variable	Observations	Mean	Std. Dev.	25th Pct.	Median	75th Pct.
Commuters	689,550	3.658	8.183	1.000	1.000	3.000
Physical Distance (km)	689,550	10.935	7.072	5.436	9.837	15.246
Travel Time (Public, min)	689,550	55.980	27.829	36.000	53.000	72.000
Travel Time (Car, min)	689,550	20.245	9.927	13.000	19.000	27.000

Notes: This table presents descriptive statistics for key commuting variables. The unit of measurement for physical distance is kilometers, while travel times are expressed in minutes. Percentiles are reported for the 25th, 50th (median), and 75th percentiles.

The summary statistics reveal several noteworthy insights into the commuting patterns within New York City. Notably, the median number of commuters per tract pair is low (1 commuter), indicating that most neighborhood pairs have minimal commuting flows. In contrast, physical distances between home and work tracts exhibit substantial variability, with a mean of approximately 10.94 kilometers and a standard deviation of 7.07 kilometers.

A particularly interesting finding is the significant difference in travel times between public transport and private vehicles. On average, commuting by public transport takes nearly 56 minutes with considerable variability (standard deviation of 27.83 minutes), whereas traveling by car averages just over 20 minutes with much less variability (standard deviation of 9.93 minutes). Differences in travel times between public and private transportation affect individuals' transportation choices and commuting expenses.

1.2 The 15-minute city

The *15-minute city* is a concept in urban planning that aims to ensure residents can access essential services and workplaces within a 15-minute commute from their homes. To evaluate this concept in the context of New York City, we computed the fraction of commuters whose commute from home to work is 15 minutes or less using the fastest available transportation mode. Our analysis reveals that 34.82% of NYC commuters meet this criterion.

Fraction of commuters with <= 15 min commute: 34.82%

Chapter 2

County-level commuting matrix

New York City is composed of five counties (Bronx, Kings, New York, Queens, and Richmond). We aggregated the tract-level commuting counts (based on the first 5 digits of each tract's 11-digit FIPS code) to obtain the county-to-county commuting flows. This information is presented in two visualizations below: a Sankey diagram (Figure 2.1) and a heatmap (Figure 2.2). Finally, the aggregated flows are saved in the file `nyc_commuting_counties.csv` containing three columns: `home_county`, `work_county`, and `commuters`.

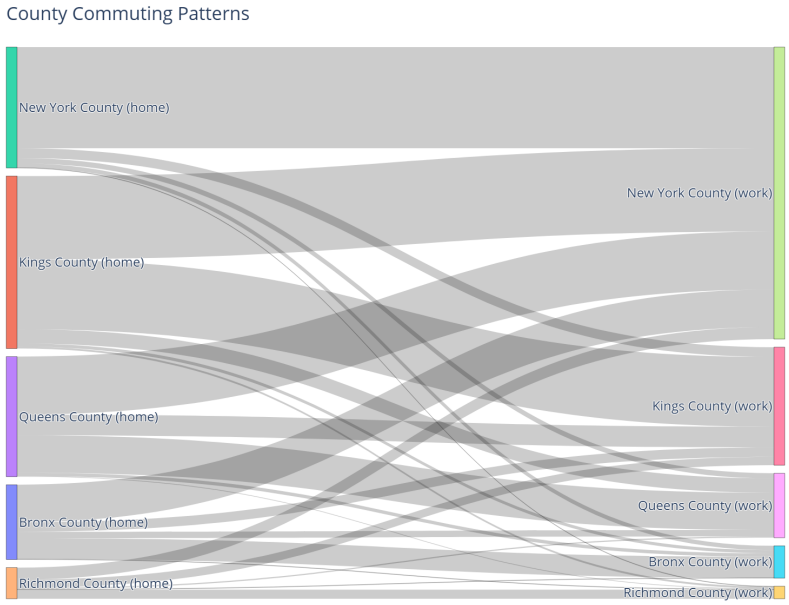


Figure 2.1: Sankey diagram showing county-to-county commuting patterns in New York City. Thicker bands indicate higher numbers of commuters.

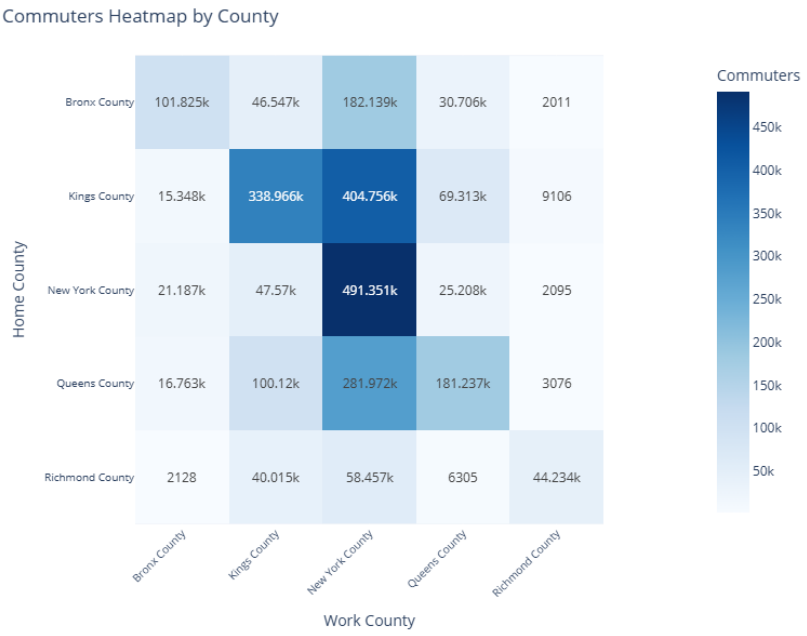


Figure 2.2: Heatmap of county-to-county commuting flows. The darker cells indicate higher volumes of commuters.

Table 2.1 displays the same county-to-county flows in numeric form:

Table 2.1: County-Level Commuting Flows in NYC

Work County	Bronx	Kings	New York	Queens	Richmond
Bronx	101825	46547	182139	30706	2011
Kings	15348	338966	404756	69313	9106
New York	21187	47570	491351	25208	2095
Queens	16763	100120	281972	181237	3076
Richmond	2128	40015	58457	6305	44234

These data highlight several major commute patterns. For instance, there are significant commuter flows from Kings County to New York County (over 400k) and substantial intra-county commuting within New York County itself (over 491k). All flows between the five counties are stored in `nyc_commuting_counties.csv` for further analysis.

Chapter 3

Regressions

3.1 Binned Scatterplot

The following analysis examines the relationship between the log of the distance between tracts and the log of the number of commuters. The scatterplot in Figure 3.1 shows the log-transformed values for both distance and commuters, with the data grouped into 100 bins based on the distance between the two tracts. This approach is useful for visualizing large datasets, as it condenses the information into binned averages, each representing the mean values of log distance and log commuters for a subset of observations.

The line of best fit is included to highlight the general trend between log distance and log commuters. Both a linear fit (solid blue line) and a cubic fit (dashed green line) are shown to capture potential nonlinear relationships. From the plot, we observe that as the log of the distance increases, the log of commuters decreases, indicating a negative relationship between the two variables.

This scatterplot confirms that commuting patterns are strongly influenced by the distance between tracts, with further distances typically correlating with fewer commuters. The choice of binning helps to better illustrate these trends while mitigating the noise that would result from plotting individual data points.

3.2 Regressions

The following table presents the results of three gravity model regressions examining the factors influencing the log number of commuters between tracts. The dependent variable in all models is the log of the number of commuters.

- **Model 1** regresses the log number of commuters on the log travel time by public transit between the tracts, showing a significant negative relationship with travel time.

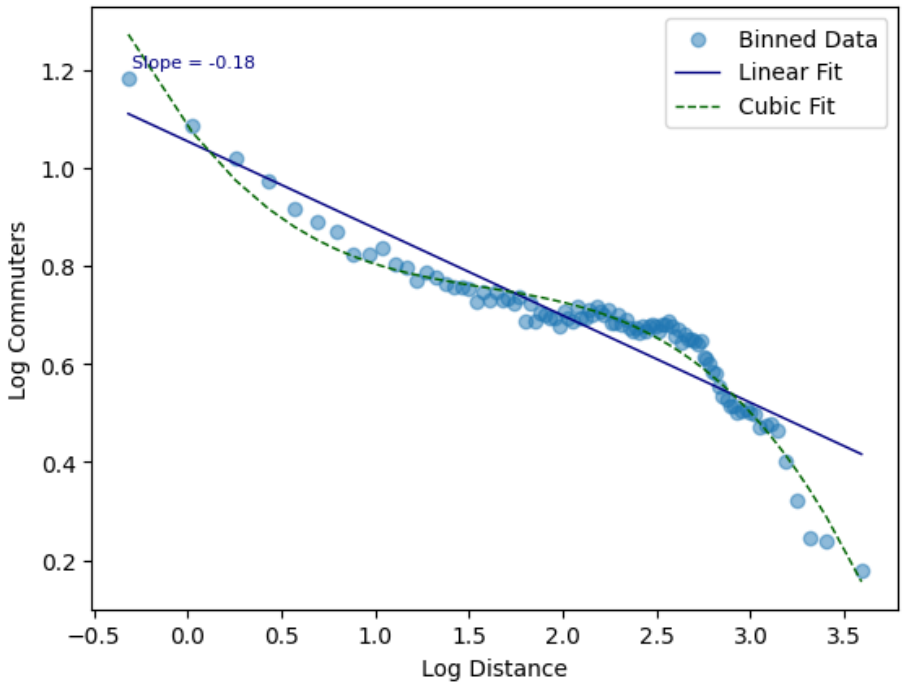


Figure 3.1: Binned Scatterplot of Log Distance vs Log Commuters

- **Model 2** introduces fixed effects for both the home tract and the work tract. This model also includes the log travel time by public transit, further refining the estimates by accounting for unobserved tract-level heterogeneity. The negative relationship between log travel time and log number of commuters remains significant.
- **Model 3** builds on Model 2 by adding the log travel time by car between the tracts. This model reveals a positive and statistically significant relationship between travel time by car and the log number of commuters.

The coefficients are robust to the inclusion of tract fixed effects and support the interpretation that travel time, both by public transit and by car, is a key determinant of commuting patterns between tracts. The results are summarized in Table 3.1.

3.3 Questions

1. Describe the relationship documented by the first regression and discuss the economic mechanisms that might explain the finding. In Model 1, the coefficient of interest on log(travel_time_public) is -0.411, indicating a negative and statistically significant relationship

Table 3.1: Regression Results

	Dependent variable: log_commuters		
	Model 1	Model 2	Model 3
	(1)	(2)	(3)
log_traveltime_public	-0.411***	-0.454***	-1.037***
	(0.002)	(0.002)	(0.004)
log_traveltime_car			0.605***
			(0.004)
home_tract FE	No	Yes	Yes
work_tract FE	No	Yes	Yes
Observations	687482	687482	687482
N. of groups	2110	2110	2110
R^2	0.068	0.068	0.106
Residual Std. Error	0.234	0.231	0.288
F Statistic	50112.854***	50054.182***	40602.709***
Note:			*p<0.1; **p<0.05; ***p<0.01
			Standard errors are in parentheses.

between public-transit travel time and the number of commuters (in logs). Economically, this suggests that longer travel times in public transport tend to reduce the volume of commuters between the origin and destination pairs, both for public transit riders and for users of private vehicles. A plausible mechanism underlying this relationship is that longer travel times increase the 'cost' (in terms of time or inconvenience) of using public transit, thus discouraging people from commuting along routes with extended durations. The corresponding negative elasticity may also reflect broader determinants, such as congestion, geographic distance, labor market matching frictions, and opportunity costs—all of which render lengthier commutes less appealing. Moreover, closer proximity between residence and workplace can alleviate these impediments by reducing commuting costs.

2. Describe how the first regression relates to the data visualization in the binned scatterplot. The binned scatterplot provides a clear visual representation of the negative relationship between log distance and log commuters, which is consistent with the results from the first regression model. As the distance between tracts increases, the number of commuters decreases, as evidenced by the downward-sloping trend in the scatterplot. This pattern mirrors the negative coefficient for log_traveltime_public in the regression, indicating that longer travel times are associated with fewer commuters. The binned scatterplot smooths the data by averaging the values within each bin, making the inverse relationship between distance and commuting volume more apparent. The slope of the best-fit line in the scatterplot, approximately -0.18, aligns closely with the regression coefficient, reinforcing the negative association. This pattern is consistent with the expectation that

longer travel times to work, which are directly related to greater distances, lead to fewer commuters.

3. Discuss why the estimated coefficient on log travel time differs between the first and second regression specifications. The coefficient on log(traveltime_public) changes from -0.411 to -0.454 between Model 1 and Model 2. The key difference is that Model 2 includes both home-tract fixed effects and work-tract fixed effects. These fixed effects control for all time-invariant differences across origin tracts and destination tracts (e.g., population density, income levels, neighborhood characteristics) that could be correlated with both travel times and commuting volumes. By purging these unobserved (and potentially confounding) factors, the second specification likely reduces omitted variable bias, causing a shift (in this case, a slightly more negative coefficient) in the estimated relationship.

4. Describe how to interpret the two travel-time coefficients in the third regression specification. In Model 3, both log(traveltime_public) and log(traveltime_car) are included as regressors, with coefficients -1.037 and $+0.605$, respectively. These coefficients are partial elasticities, meaning each coefficient measures the percentage change in log(commuters) given a 1% change in the respective travel-time measure, holding the other travel mode's travel time constant. A 1% increase in public-transit travel time is associated with a 1.037% decrease in the number of commuters, all else equal; conversely, a 1% increase in car travel time is associated with a 0.605% *increase* in the number of commuters. This suggests that, with fixed public transport travel times and fixed origin and destination effects, longer car travel times are associated with a higher number of commuters. Given that the decision to expand public transport is influenced by commuter numbers, this model may be subject to endogeneity and requires further analysis to make causal inferences.

5. Why is it infeasible to create a same-tract dummy (for home tract = work tract) and add it to the regression? Once we include both home-tract and work-tract fixed effects, the model cannot separately identify the effect of a dummy variable that is 1 if and only if the home tract equals the work tract. This dummy is **perfectly collinear** with the combination of the home-tract and work-tract fixed effects, because each origin tract and each destination tract already has its own indicator. In other words, for any given tract, the case where it is both the home and work tract is a single observation perfectly determined by those two fixed effects. Hence, adding a same-tract indicator does not add new variation beyond what is already captured by the fixed effects, making it infeasible for the regression to estimate a separate coefficient on that dummy.

References

Angrist, J. (2014). *Mastering'metrics: The path from cause to effect*. Princeton University Press.

Wooldridge, J. M., Wadud, M., & Lye, J. (2016). *Introductory econometrics: Asia pacific edition with online study tools 12 months*. Cengage AU.

Appendix A

Python Code

A.1 Introduction

This appendix presents selected Python scripts integral to the analysis of commuting flows between neighborhoods in New York City. The complete codebase is available in the accompanying repository, which encompasses the following key functionalities:

1. **Summary Statistics**: Generates comprehensive summary statistics tables for tract-to-tract variables, facilitating an initial understanding of the dataset.
2. **Commuting Dataset Construction**: Constructs a county-to-county commuting dataset and provides visual representations to elucidate commuting patterns.
3. **Binned Scatterplot Generation**: Creates binned scatterplots with the logarithm of distance on the x-axis and the logarithm of the number of commuters on the y-axis, aiding in the visualization of spatial relationships.
4. **Gravity-Model Regressions**: Reports estimates derived from three distinct gravity-model regressions, offering insights into the factors influencing commuting behaviors.

The scripts are implemented in Python and are executable both on Google Colab and local environments. Execution via Google Colab is recommended due to the streamlined installation of required libraries using `pip`. For local execution, ensure that all dependencies listed in the `Requirements` part are installed prior to running the code.

A.2 Access Instructions

- *Google Colab*: https://colab.research.google.com/drive/14w0kvylNAy5p2aApdTvbRsVuoZT7VB1z?usp=sharing

- An active internet connection is necessary as all necessary files are fetched online; no additional local files need to be loaded.

A.3 Snippets

A.3.1 Summary Statistics Table

The following Python function generates a summary statistics table for the relevant variables, which is subsequently exported to a LaTeX file for inclusion in the report.

Summary Statistics Table

```
summary_stats = data.drop(columns=['home_tract',
↪  'work_tract']).describe(percentiles=[0.25, 0.5, 0.75]).transpose()
summary_stats = summary_stats[['count', 'mean', 'std', '25%', '50%', '75%']]
summary_stats['count'] = summary_stats['count'].apply(lambda x: "%d" % x)
summary_stats.iloc[:, 1:] = summary_stats.iloc[:, 1:].applymap(lambda x: "%.3f"
↪  % x)

print(summary_stats)
# Escape underscore, and percent sign
summary_stats.index = summary_stats.index.str.replace(r'([%_])', r'\\\1',
↪  regex=True)
summary_stats.columns = summary_stats.columns.str.replace('%', '\%',
↪  regex=False)

# Export to LaTeX
latex_code = summary_stats.to_latex(
    index=True,
    caption="Summary Statistics",
    label="tab:summary_stats",
)

# Save to file
with open('summary_stats.tex', 'w') as f:
    f.write(latex_code)

print("Summary statistics exported to LaTeX file successfully.")
```

A.3.2 The 15-minute city

The following code determines each commuter's shortest travel time by selecting the minimum between public transportation and car travel times, and subsequently calculates the percentage of commuters whose fastest commute does not exceed fifteen minutes.

The 15-minute city

```
data['fastest_time'] = data[['traveltime_public', 'traveltime_car']].min(axis=1)
fraction_15_minute_commuters = (data['fastest_time'] <= 15).sum() / len(data)
print(f"Fraction of commuters with <= 15 min commute:
↪  {fraction_15_minute_commuters:.2%}")
```

A.3.3 Binned Scatterplot

The following code constructs a binned scatterplot to examine the relationship between the logarithm of physical distance and the logarithm of the number of commuters. It performs log transformations, aggregates the data into quantile-based bins, applies linear and cubic regression fits, and visualizes the results with corresponding annotations.

Binned Scatterplot

```
data['log_distance'] = np.log(data['physical_distance'])
data['log_commuters'] = np.log(data['commuters'])

bins = pd.qcut(data['physical_distance'], 100, duplicates='drop')
binned_data = data.groupby(bins).agg({'log_distance': 'mean', 'log_commuters':
↪  'mean'}).reset_index(drop=True)
binned_data.replace([np.inf, -np.inf], np.nan, inplace=True)
binned_data.dropna(inplace=True)

plt.scatter(
    binned_data['log_distance'],
    binned_data['log_commuters'],
    alpha=0.5,
    label='Binned Data'  # Add legend label for data points
)
```

Linear Fit

```
slope, intercept, _, _, _ = linregress(binned_data['log_distance'],
↪  binned_data['log_commuters'])
plt.plot(
    binned_data['log_distance'],
    intercept + slope * binned_data['log_distance'],
    color='navy',
    linewidth=1,
    label='Linear Fit'
)
```

Annotate the slope

```
plt.annotate(
    f"Slope = {slope:.2f}",       # Text to display
    xy=(0.05, 0.9),              # Position in Axes coordinates (0=left/bottom,
    ↪  1=right/top)
    xycoords='axes fraction',    # Interpret (x,y) as a fraction of the Axes
    fontsize=8,
    color='navy'
)
```

Polynomial (Cubic) Fit

```
coeffs = np.polyfit(binned_data['log_distance'], binned_data['log_commuters'],
↪  3)
cubic_fit = np.poly1d(coeffs)
plt.plot(
    binned_data['log_distance'],
    cubic_fit(binned_data['log_distance']),
    color='darkgreen',
    linestyle='--',
    linewidth=1,
    label='Cubic Fit'
)
```

binsreg

```
# Use binsreg to create a non-parametric binned scatterplot
data_fe = data.copy()
data_fe.replace([np.inf, -np.inf], np.nan, inplace=True)
data_fe.dropna(inplace=True)
est = binsreg.binsreg(data_fe['log_commuters'], data_fe['log_distance'],
↪  nbins=100, polyreg=1)
est.bins_plot.show()
```

A.3.4 Regressions

The following code performs data preparation by log-transforming travel time variables, executes three regression models (including pooled OLS and fixed effects), and generates a LaTeX table of the regression results using the Stargazer package.

Regressions

```
# Data Preparation for Regression
data_fe['log_traveltime_public'] = np.log(data_fe['traveltime_public'])
data_fe['log_traveltime_car'] = np.log(data_fe['traveltime_car'])

# Regress log commuters on log travel time by public transit
data_fe = data_fe.set_index(['home_tract', 'work_tract'])
X1 = data_fe[['log_traveltime_public']]
y = data_fe['log_commuters']
model1 = panel.PooledOLS(y, sm.add_constant(X1)).fit()

# Fixed effects regression with home and work tract
model2 = panel.PanelOLS(y, sm.add_constant(X1), entity_effects=True).fit()

# Add log travel time by car and fixed effects
X3 = data_fe[['log_traveltime_public', 'log_traveltime_car']]
model3 = panel.PanelOLS(y, sm.add_constant(X3), entity_effects=True).fit()

# Use Stargazer to generate LaTeX tables
stargazer = Stargazer([model1, model2, model3])
stargazer.title('Regression Results')
stargazer.custom_columns(['Model 1', 'Model 2', 'Model 3'], [1, 1, 1])
```

```
stargazer.significance_levels([0.1, 0.05, 0.01])
stargazer.show_degrees_of_freedom(False)
stargazer.covariate_order(['log_traveltime_public', 'log_traveltime_car'])
stargazer.add_line('home_tract FE', ['No', 'Yes', 'Yes'])
stargazer.add_line('work_tract FE', ['No', 'Yes', 'Yes'])
stargazer.add_custom_notes(["Standard errors are in parentheses."])

latex_output = stargazer.render_latex()

# Save to a .tex file
with open("regression_results.tex", "w") as f:
    f.write(latex_output)
```

Appendix B

County-to-County Commuting Table

home_county	work_county	commuters
36005	36005	101825
36005	36047	46547
36005	36061	182139
36005	36081	30706
36005	36085	2011
36047	36005	15348
36047	36047	338966
36047	36061	404756
36047	36081	69313
36047	36085	9106
36061	36005	21187
36061	36047	47570
36061	36061	491351
36061	36081	25208
36061	36085	2095
36081	36005	16763
36081	36047	100120
36081	36061	281972
36081	36081	181237
36081	36085	3076
36085	36005	2128
36085	36047	40015
36085	36061	58457
36085	36081	6305
36085	36085	44234

Table B.1: Commuters by Home and Work County